The cylinders have steel cylinder walls that expand slowly compared to aluminum pistons that expand quickly. If an engine is revved too quickly during takeoff before warming up, a lot of heat is generated on top of the piston. That quickly expands the piston, which can then seize in the cylinder. A piston seizure will stop the engine abruptly.

The second area of concern is lower in the engine around the engine crankshaft. This is an area where things may get too loose with heat, rather than seizing up. Additionally, the crankcase has steel bearings set into the aluminum which need to expand together or the bearings could slip.

Many two-stroke engines have steel bearings that normally hug the walls of the aluminum engine case. The crank spins within the donuts of those steel bearings. If you heat up the engine two quickly, the aluminum case will out-expand those steel bearings and the crank will cause the bearings to start spinning along with it. If those steel bearings start spinning, they can ruin the soft aluminum walls of the case, which is very expensive.

If heat is slowly added to an engine, all the parts will expand more evenly. This is done through a proper warm-up procedure. Many two-stroke engines are best warmed up by running the engine at a set RPM for a set amount of time. Follow the instructions in your POH; however, a good rule of thumb is to initially start the engine at idle RPM, get it operating smoothly, and then warm the engine at 3,000 RPM for 5 minutes.

Once the engine is warmed up and the powered parachute is flying, it is still possible to cool down the engine too much. This will happen when the engine is idled back for an extended period of time. Even though the engine is running, it is not generating as much heat as the cooling system is efficiently dumping into the atmosphere. An immediate power application with a cooled engine can seize the engine just as if the engine had not been warmed in the first place.

In water-cooled engines, on a long descent at idle, the coolant cools until the thermostat closes and the engine is not circulating the radiator fluid through the engine. The engine temperature remains at this thermostat closed temperature while the radiator coolant continues to cool further. If full throttle is applied, the thermostat can open allowing a blast of coolant into the warm engine. The piston is expanding because of the added heat and the cylinder is cooling with the cold radiator water resulting in a piston seizure. To prevent this, slowly add power well before you get close to the ground where you will need power. This will give the system a chance to gradually open the thermostat and warm up the radiator water.

Just as it takes a while for the engine crankcase and bearings to warm up, it also takes those steel parts a long time to cool down. If you land, refuel and want to take off again quickly, there is no need to warm up again for 5 minutes. The lower end of the engine will stay warmed up after being shut down for short periods. An engine restart is an example where it would be appropriate to warm the engine up until the gauges reach operating temperatures. The lower end of the engine is warm and now you only need to be concerned with preventing the pistons from seizing.

Four-Stroke Engine Warming

A four-stroke engine must also be warmed up. The four-stroke engine has a pressurized oil system that provides more uniform engine temperatures to all its components. You can apply takeoff power as soon as the water, cylinder head temperature (CHT), oil temperatures and oil pressure are within the manufacturer's recommended tolerances for takeoff power applications.

Gearboxes

Gearboxes are used on all powered parachute reciprocating engines to take the rotational output of an internal combustion engine which is turning at a very high RPM and convert it to a slower (and more useful) RPM to turn the propeller. Gearboxes come in different gear ratios depending on the output speed of the engine and the needed propeller turning speeds.

A typical two-stroke RPM reduction is from 6,500 engine RPM with a 3.47 to 1 reduction, resulting in 1,873 propeller RPM. A typical four-stroke RPM reduction is from 5,500 engine RPM with a 2.43 to 1 reduction, resulting in 2,263 propeller RPM.

A gearbox is a simple device that bolts directly to the engine and in turn has the propeller bolted directly to it.

A two-cycle engine gearbox is kept lubricated with its own built-in reservoir of heavy gearbox oil. The reservoir is actually part of the gearbox case itself. The gearbox oil has to be changed periodically since the meshing of the gears will cause them to wear and will deposit steel filings into the oil. If the oil is not